

4

Complex Integration

4.1 Introduction

We are already familiar with the definite integral of a function of a real variable. This chapter deals with the integral of a function of a complex variable. An integral of a function of a complex variable over a curve c in the complex plane is called the complex line integral. It is denoted by

$$\int_c f(z)dz,$$

where $f(z)$ is a function of a complex variable and c is the path of integration.

4.2 Integral as the Limit of a Sum

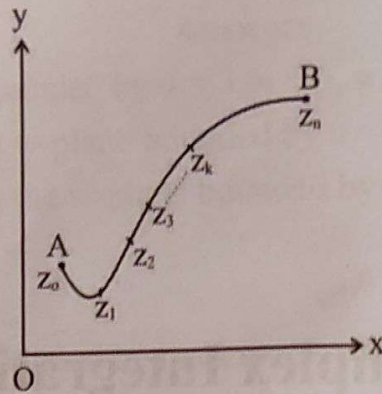
Let c be a smooth curve defined by $z = x(t) + i y(t)$, ($a \leq t \leq b$). Divide the interval $[a, b]$ by the points

$$a = t_0, t_1, t_2, t_3, \dots, t_n = b$$

and let the corresponding subdivision of c be

$$A = z_0, z_1, z_2, z_3, \dots, z_n$$

Form the sum $\sum f(z_k)\Delta z_k$; where $\Delta z_k = z_k - z_{k-1}$



If the limit of this sum taken over all possible partitions tends to a unique limit as $n \rightarrow \infty$ and the greatest of $\Delta z_k \rightarrow 0$, we say that $f(z)$ is integrable over c and the limit is called the line integral of $f(z)$ over the curve c . The line integral is denoted by $\int_c f(z) dz$.

4.3 Reduction to the Line Integrals of Real Functions

Let $f(z) = u + iv$, where $z = x + iy$. Then

$$\int_c f(z) dz = \int_c (u + iv) (dx + idy) = \int_c (u dx - v dy) + i \int_c (v dx + u dy)$$

It shows that the line integral of a complex function can be evaluated by reducing it to the line integrals of the real functions.

Further, if the curve has a parametric representation:

$$z = z(t) = x(t) + iy(t) \quad (a \leq t \leq b)$$

$$\text{then } \int_c f(z) dz = \int_a^b f[z(t)] z'(t) dt.$$

An Important Theorem

Let $f(z)$ be analytic in domain D . If there exists an analytic function $F(z)$ such that $F'(z) = f(z)$, then

$$\int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0)$$

For example, $\int_0^{1+i} z dz = \left. \frac{z^2}{2} \right|_0^{1+i} = i$

Some Properties of Complex Integrals

1. $\int_c [f(z) + \phi(z)] dz = \int_c f(z) dz + \int_c \phi(z) dz$
2. $\int_c k f(z) dz = k \int_c f(z) dz$, where k is a complex constant.
3. Let k_1 and k_2 be two complex constants, then

$$\int_c [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_c f_1(z) dz + k_2 \int_c f_2(z) dz$$
4. $\int_{z_2}^{z_1} f(z) dz = - \int_{z_1}^{z_2} f(z) dz$ (sense reversed)
5. If c is composed of two parts c_1 and c_2 , then

$$\int_c f(z) dz = \int_{c_1} f(z) dz + \int_{c_2} f(z) dz$$

The proofs of the above properties follow from the definition of a complex integral.

Examples

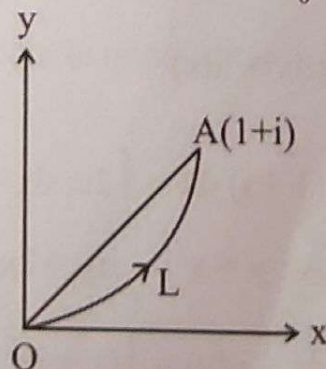
1. Integrate $\int_c \operatorname{Re} z dz$ from 0 to $1 + i$ along the path c , where
 (a) c : a st. line $y = x$ (b) c : a parabola $y = x^2$

Let $z = x + iy$ so that $dz = dx + idy$

$$\int_c \operatorname{Re} z dz = \int_c x (dx + idy) \quad (1)$$

- a. Along the path c ; $y = x$: $dy = dx$.

$$\therefore \int_c \operatorname{Re} z dz = \int_c x (dx + idx) = (1 + i) \int_0^1 x dx = \frac{1}{2} (1 + i)$$



- b. Along the path c , $y = x^2$: $dy = 2x dx$

$$\begin{aligned}\int_c \operatorname{Re} z \, dz &= \int_c x (dx + i dy) = \int_c x (dx + i 2x dx) \\ &= \int_0^1 x (1 + 2ix) \, dx = \int_0^1 (x + 2ix^2) \, dx = \frac{1}{2} + \frac{2}{3} i\end{aligned}$$

Thus, $\int_c \operatorname{Re} z \, dz$ along the line $y = x$ is $\frac{1}{2}(1 + i)$, and

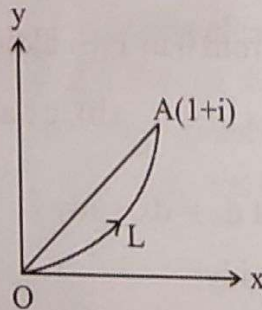
$$\int_c \operatorname{Re} z \, dz \text{ along the curve } y = x^2 \text{ is } \frac{1}{2} + \frac{2}{3} i.$$

It shows that the integral depends on path. It is so because $\operatorname{Re} z$ is not analytic. However, if $f(z)$ is analytic, the integral will not depend on the path. Consider the following example.

2. Integrate $\int_c z \, dz$, from 0 to $1 + i$ along the path c , where

(a) c : a st. line $y = x$

(b) a curve $y = x^2$



- a. Along the path OA : $y = x$, $dy = dx$

$$\begin{aligned}\int_c z \, dz &= \int_c (x + iy) (dx + i dy) \\ &= \int_c (x + ix) (dx + i dx) \\ &= \int_0^1 (x + ix) (1 + i) \, dx = \int_0^1 2ix \, dx = i\end{aligned}$$

- b. Along the path OLA : $y = x^2$, $dy = 2x dx$

$$\int_c z \, dz = \int_c (x + iy) (dx + i dy) = \int_0^1 (x + ix^2) (1 + 2ix) \, dx$$

$$= \int_0^1 (x - 2x^3 + 3ix^2) dx = i$$

Thus, along the path OA, $\int_C z dz = i$, and

along the path OLA, $y = x^2$, $\int_C z dz = i$

The values of the integral in different paths are all equal. It hints that if the function is analytic, the integral does not depend on the path traversed. Moreover, the integral of z round the closed contour is

$$\int_{OLAO} z dz = \int_{OLA} z dz + \int_{AO} z dz = \int_{OLA} z dz - \int_{OA} z dz = i - i = 0$$

We note that z is analytic and its integral around the closed contour is zero. In the next section, we shall prove a general result known as Cauchy's integral theorem.

3. Integrate $\int_C \frac{1}{z} dz$, counter clockwise around the circle $|z| = 1$.

The equation of the circle is $|z| = 1$. In parametric form,

$$z = \cos t + i \sin t = e^{it} \quad (0 \leq t \leq 2\pi)$$

so that $dz = ie^{it} dt$. Now,

$$\int_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} \cdot ie^{it} dt = i \int_0^{2\pi} dt = 2\pi i$$

4. Integrate $\int_C (z - z_0)^n$, where n is an integer, z_0 a complex constant, around the circle with centre z_0 and radius r .

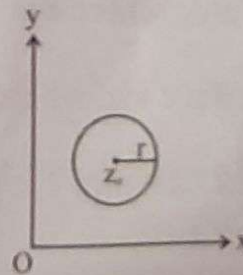
The equation of a circle with centre z_0 and radius r is given by

$$|z - z_0| = r$$

In parametric form,

$$z - z_0 = re^{it}$$

$$z = z_0 + re^{it} \quad (0 \leq t \leq 2\pi)$$



$$\therefore dz = ire^{it}$$

$$\text{Now, } \int_c (z - z_0)^n dz = \int_0^{2\pi} (re^{it})^n \cdot ire^{it} dt = ir^{n+1} \int_0^{2\pi} e^{it(n+1)} dt$$

$$= ir^{n+1} \int_0^{2\pi} e^{it(n+1)} dt \quad (1)$$

$$= \frac{ir^{n+1}}{i(n+1)} \cdot e^{i(n+1)t} \Big|_0^{2\pi}$$

$$= \frac{r^{n+1}}{n+1} [e^{i(n+1)2\pi} - 1]$$

$$= \frac{r^{n+1}}{n+1} [\cos 2(n+1)\pi + i \sin 2(n+1)\pi - 1]$$

If $n \neq -1$, then $\cos 2(n+1)\pi = 1$ and $\sin 2(n+1)\pi = 0$

$$\therefore \int_c (z - z_0)^n dz = 0 \text{ when } n \neq -1.$$

$$\text{If } n = -1, \text{ then } \int_c (z - z_0)^n dz = i \int_0^{2\pi} dt = 2\pi i.$$

Thus,

$$\int_c (z - z_0)^n dz = \begin{cases} 2\pi i & \text{for } n = -1 \\ 0 & \text{for } n \neq -1 \end{cases}$$

Exercise 4.1

Integrate the following integrals along the given path

- $\int_c z^2 dz$ where c is a shortest path joining $1+i$ and $2(1+2i)$
- $\int_c \operatorname{Re}(z^2) dz$, where c is an unit circle counterclockwise.
- $\int_c \bar{z} dz$ where c is a curve from 0 along the parabola $y = x^2$ to $1+i$

- iv. $\int_c (12z^2 - 4iz) dz$ along the curve joining the points (1, 1) and (2, 3)
v. $\int_c \log z dz$ where c is unit circle $|z| = 1$
vi. $\int_c \cos z dz$ where c is semicircle $|z| = \pi$ $x \geq 0$ from $-\pi i$ to πi
vii. $\int_c \sin^2 z dz$, where c is from $-\pi i$ to πi in right half plane

Answers

- i. $-\frac{86}{3} - 6i$ ii. 0
iii. $1 + \frac{i}{3}$ iv. $-156 + 38i$
v. $2\pi i$ vi. $2i \sinh \pi$
vii. $(\pi + \frac{\sinh 2\pi}{2})$

4.4 Path and Domain

A closed path is called a simple closed path if it does not intersect or touch itself. For example, a square is a simple closed path, but an 8-shaped curve is not.



simple



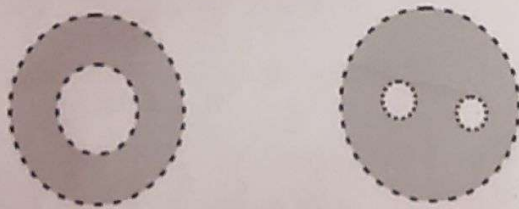
not simple

A simple closed path is also called contour and an integral over such a path is called **contour integral**.

A domain is called **simply connected** if every closed path in the domain encloses the points of the domain only. For example, the interior of a circle, or an ellipse or a square is simply connected.



A domain that is not simply connected is called **multiply connected**. For example, the domain between two concentric circles is a multiply connected domain. It gives birth to holes. In particular, a domain with one hole is called doubly connected and a domain with two holes is called triply connected.



4.5 Fundamental Theorem of Function Theory

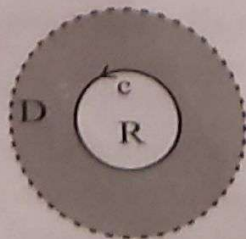
Theorem 1. (Cauchy's Integral Theorem)

If $f(z)$ is analytic in a simply connected domain D , and $f'(z)$ is continuous at each point of D , then for every closed path c in D ,

$$\int_c f(z) dz = 0$$

Proof.

Let $f(z)$ be analytic in a simply connected domain D , and let R be a closed region whose boundary is c .



Let $f(z) = u + iv$, and $z = x + iy$ so that $dz = dx + idy$.

$$\int_c f(z) dz = \int_c (u + iv) (dx + idy) = \int_c (u dx - v dy) + i \int_c (v dx + u dy)$$

Since $f(z)$ is analytic in D , so $f'(z)$ exists. Since $f(z)$ be continuous, so u, v and their partial derivatives are

Using Green's theorem, we have

$$\int_c (u dx - v dy) = \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy \quad (1)$$

$$\int_c (v dx + u dy) = \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy \quad (2)$$

Since $f(z)$ is analytic, so $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

The integrals on the right side of (1) and (2) are each

$$\therefore \int_c f(z) dz = 0.$$

This proof was provided by Cauchy (1789-1857) in which $f'(z)$ is assumed to be continuous. Later proved the theorem without assuming the continuity. The proof given by Goursat is more complicated, and we do not consider it.

Examples

1. Analytic Functions

We know that the functions

$$e^z, \sin z, \cos z, z^n \quad (n = 0, 1, 2, \dots)$$

are analytic for all z . So,

$$\int_c e^z dz = 0, \quad \int_c \sin z dz = 0, \quad \int_c \cos z dz = 0$$

where c is any closed path.

2. Not analytic outside the closed path

$$i) \quad \int_c \frac{1}{z^2 + 9} dz, \quad \text{where } c : |z| = 1$$

Let $f(z) = u + iv$, and $z = x + iy$ so that $dz = dx + idy$. Now,

$$\int_C f(z) dz = \int_C (u + iv) (dx + idy) = \int_C (udx - vdy) + i \int_C (vdx + udy)$$

Since $f(z)$ is analytic in D , so $f'(z)$ exists. Since $f'(z)$ is assumed to be continuous, so u , v and their partial derivatives are continuous.

Using Green's theorem, we have

$$\int_C (udx - vdy) = \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy \quad (1)$$

$$\int_C (vdx + udy) = \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy \quad (2)$$

Since $f(z)$ is analytic, so $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

The integrals on the right side of (1) and (2) are each zero.

$$\therefore \int_C f(z) dz = 0.$$

This proof was provided by Cauchy (1789-1857) in the year 1814 in which $f'(z)$ is assumed to be continuous. Later on Goursat proved the theorem without assuming the continuity of $f'(z)$. The proof given by Goursat is more complicated, and therefore we do not consider it.

Examples

1. Analytic Functions

We know that the functions

$$e^z, \sin z, \cos z, z^n \quad (n = 0, 1, 2, \dots)$$

are analytic for all z . So,

$$\int_C e^z dz = 0, \quad \int_C \sin z dz = 0, \quad \int_C \cos z dz = 0, \quad \int_C z^n dz = 0$$

where C is any closed path.

2. Not analytic outside the closed path

$$i) \quad \int_C \frac{1}{z^2 + 9} dz, \quad \text{where } C : |z| = 1$$

The integrand is not analytic at the points $\pm 3i$, but these points lie outside the circle $|z| = 1$. That is, the function is analytic within and on the circle $|z| = 1$. So,

$$\int_c \frac{1}{z^2 + 9} dz = 0$$

ii) $\int_c \tan z \, dz, \quad c : |z| = 1$

$\tan z$ is not analytic at $\pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$, but all these lie outside $c : |z| = 1$. So,

$$\int_c \tan z \, dz = 0$$

3. Analytic sufficient, not necessary

$$\int_c \frac{1}{z^2} dz, \quad c : |z| = 1$$

The function $\frac{1}{z^2}$ is not analytic at $z = 0$, which lies within the circle $|z| = 1$. So, we cannot apply Cauchy's integral theorem. We proceed as follows:

Set $z = e^{it}$ so that $dz = ie^{it} dt$

$$\int_c \frac{1}{z^2} dz = \int_0^{2\pi} \frac{1}{e^{2it}} ie^{it} dt = i \int_0^{2\pi} e^{-it} dt = -e^{-it} \Big|_0^{2\pi} = -(e^{-i2\pi} - 1)$$

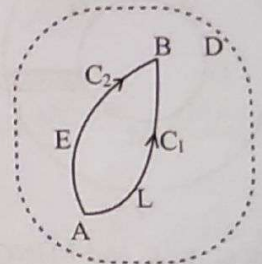
$$= -[\cos 2\pi - i \sin 2\pi - 1] = 0$$

It shows that the integral of a function may be zero although it is not analytic. Hence the condition that $f(z)$ be analytic in D is sufficient rather than necessary.

Theorem 2. (Independence of Path)

If $f(z)$ is analytic in a simply connected domain D , then the integral of $f(z)$ is independent of path in D .

Proof.



Let ALB represent the curve C_1 , and ALBEA is a simple closed path. Since $f(z)$ is analytic in D , by Cauchy's integral theorem,

$$\int_{ALBEA} f(z) dz = 0$$

$$\text{or, } \int_{ALB} f(z) dz + \int_{BEA} f(z) dz = 0$$

$$\text{or, } \int_{ALB} f(z) dz - \int_{AEB} f(z) dz = 0$$

$$\text{or, } \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

$\Rightarrow$ The integral is independent of path.

4.6 Extension to the Multiply Connected Domain

Theorem 3. Let D be the doubly connected domain. Let C and C_1 be two simple closed curves in D such that C is the outer boundary and C_1 is the inner boundary. Then

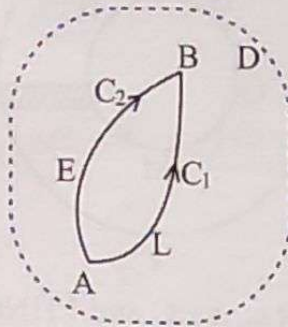
$$\int_C f(z) dz = \int_{C_1} f(z) dz$$

where C and C_1 are both traversed in the anticlockwise direction).

Theorem 2. (Independence of Path)

If $f(z)$ is analytic in a simply connected domain D , then the integral of $f(z)$ is independent of path in D .

Proof.



Let ALB represent the curve C_1 , and AEB curve C_2 . Then ALBEA is a simple closed path. Since $f(z)$ is analytic in D , so by Cauchy's integral theorem.

$$\int_{ALBEA} f(z) dz = 0$$

$$\text{or, } \int_{ALB} f(z) dz + \int_{BEA} f(z) dz = 0$$

$$\text{or, } \int_{ALB} f(z) dz - \int_{AEB} f(z) dz = 0$$

$$\text{or, } \int_{c_1} f(z) dz = \int_{c_2} f(z) dz$$

$\Rightarrow$ The integral is independent of path.

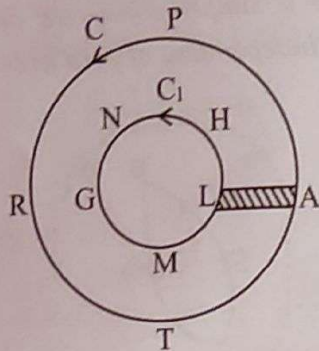
4.6 Extension to the Multiply Connected Domain

Theorem 3. *Let D be the doubly connected domain bounded by two simple closed curves C and C_1 such that C_1 is contained in C , then*

$$\oint_C f(z) dz = \oint_{C_1} f(z) dz$$

where C and C_1 are both traversed in the positive sense (i.e. anticlockwise direction).

Proof.



Let APRTA represent curve C and LHNGML C_1 .

Let us make a cross-cut joining a point A of curve C to a point L of curve C_1 as shown in the figure. Then APRTALMGNHCLA is a simple closed path. By Cauchy's theorem

$$\int_{APRTALMGNHCLA} f(z) dz = 0$$

$$\text{or, } \int_{APRTA} f(z) dz + \int_{AL} f(z) dz + \int_{LMGNHL} f(z) dz + \int_{LA} f(z) dz = 0$$

$$\text{or, } \int_{APRTA} f(z) dz = - \int_{LMGNHL} f(z) dz = \int_{LHNGML} f(z) dz$$

$$\text{or, } \int_C f(z) dz = \int_{C_1} f(z) dz$$

Similarly, if $f(z)$ is analytic in multiply connected domain bounded by the non-overlapping curves C, C_1 , C_2 where C_1 and C_2 are inside C, then

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

where C, C_1 , C_2 are traversed in the anticlockwise direction.

An Important Result

Using Theorem 3 and Example 4, (page 54) it follows that

$$\int_C (z - z_0)^n dz = \begin{cases} 2\pi i & \text{for } n = -1 \\ 0 & \text{for } n \neq -1 \end{cases} \quad (n \text{ integer})$$

the integration being taken counter clockwise over a closed path containing z_0 in its interior.

In particular,

$$\int_C \frac{1}{z - z_0} dz = 2\pi i, \quad (n = -1) \quad \int_C (z - z_0)^n dz = 0$$

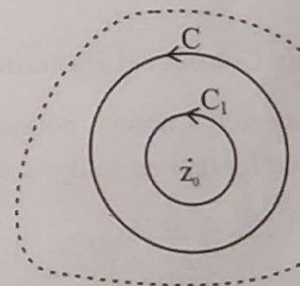
$$\int_C \frac{1}{(z - z_0)^2} dz = 0, \quad (n = -2) \quad \int_C (z - z_0)^n dz = 0$$

$$\int_C \frac{1}{(z - z_0)^3} dz = 0, \quad (n = -3) \quad \int_C (z - z_0)^n dz = 0$$

4.7 Cauchy's Integral Formula. Let $f(z)$ be analytic in a connected domain D, then for any point z_0 in D

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

where C is any simple closed path that encloses z_0 .



Proof.

Since $f(z)$ is analytic in D, so $\frac{f(z)}{z - z_0}$ is analytic in D except at the point $z = z_0$. Let C be a closed path C except at the point $z = z_0$, with centre at $z = z_0$ and radius r so that $\frac{f(z)}{z - z_0}$ is analytic in the annulus bounded by C and C1.

the integration being taken counter clockwise around any simple closed path containing z_0 in its interior.

In particular,

$$\int_C \frac{1}{z - z_0} dz = 2\pi i, \quad (n = -1) \quad \int_C (z - z_0) dz = 0 \quad (n = 1)$$

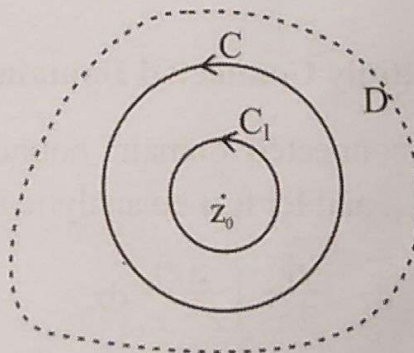
$$\int_C \frac{1}{(z - z_0)^2} dz = 0, \quad (n = -2) \quad \int_C (z - z_0)^2 dz = 0 \quad (n = 2)$$

$$\int_C \frac{1}{(z - z_0)^3} dz = 0, \quad (n = -3) \quad \int_C (z - z_0)^3 dz = 0 \quad (n = 3)$$

4.7 Cauchy's Integral Formula. Let $f(z)$ be analytic in a simply connected domain D , then for any point z_0 in D ,

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

where C is any simple closed path that encloses z_0 .



Proof.

Since $f(z)$ is analytic in D , so $\frac{f(z)}{z - z_0}$ is analytic inside and on the closed path C except at the point $z = z_0$. Describe a small circle C_1 with centre at $z = z_0$ and radius r so that C_1 lies inside C . Now, $\frac{f(z)}{z - z_0}$ is analytic in the annulus bounded by C and C_1 . Then

$$\int_C \frac{f(z)}{z - z_0} dz = \int_{C_1} \frac{f(z)}{z - z_0} dz \quad (1)$$

where C and C_1 are both taken in the counter clockwise direction.
The equation of the circle C_1 in parametric form is

$$z - z_0 = re^{it} \quad (0 \leq t \leq 2\pi)$$

$$\therefore dz = ire^{it} dt$$

$$\text{Now, } \int_{C_1} \frac{f(z)}{z - z_0} dz = \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} ire^{it} dt = i \int_0^{2\pi} f(z_0 + re^{it}) dt$$

$$\text{Let } r \rightarrow 0, \text{ then } \int_{C_1} \frac{f(z)}{z - z_0} dz = i f(z_0) \int_0^{2\pi} dt = 2\pi i f(z_0)$$

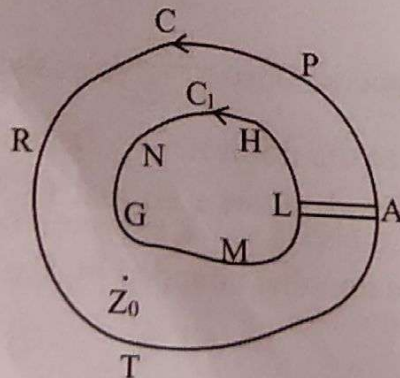
$$\therefore \int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0) \text{ i.e. } f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

The theorem thus relates the value of an analytic function at a point with the values on a closed path not containing the point.

Extension to the Multiply Connected Domain

Let D be a doubly connected domain bounded by two simple closed curves C and C_1 , and let $f(z)$ be analytic in D , then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz - \frac{1}{2\pi i} \int_{C_1} \frac{f(z)}{z - z_0} dz$$



Proof.

Let the curve C be represented by APRTA and C_1 by LHNGML. Make a cross-cut by joining A and L. Then $f(z)$ is analytic in the region bounded by APRTALMGNHILA. By Cauchy's integral formula,

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_{\text{APRTALMGNHILA}} \frac{f(z)}{z - z_0} dz \\ &= \frac{1}{2\pi i} \left[\int_{\text{APRTA}} \frac{f(z)}{z - z_0} dz + \int_{\text{AL}} \frac{f(z)}{z - z_0} dz + \int_{\text{LMGNHL}} \frac{f(z)}{z - z_0} dz + \int_{\text{LA}} \frac{f(z)}{z - z_0} dz \right] \\ &= \frac{1}{2\pi i} \left[\int_C \frac{f(z)}{z - z_0} dz - \int_{\text{LHNGML}} \frac{f(z)}{z - z_0} dz \right] \\ &= \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz - \frac{1}{2\pi i} \int_{C_1} \frac{f(z)}{z - z_0} dz \end{aligned}$$

Similarly, we can show that for triply connected domain

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz - \frac{1}{2\pi i} \int_{C_1} \frac{f(z)}{z - z_0} dz - \frac{1}{2\pi i} \int_{C_2} \frac{f(z)}{z - z_0} dz$$

4.8 Derivatives of an Analytic Function

Let $f(z)$ be analytic in a simply connected domain D . Then, for any point z_0 in D and any simple closed path that encloses z_0 ,

$$f'(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^2} dz$$

The proof of this result is beyond the scope of the book. For memorizing the result will be obtained by differentiating the

Cauchy's integral formula under the integral sign with respect to the parameter z_0 . In the same way, we have

$$f''(z_0) = \frac{2!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^3} dz$$

$$f'''(z_0) = \frac{3!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^4} dz$$

and in general, $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz$

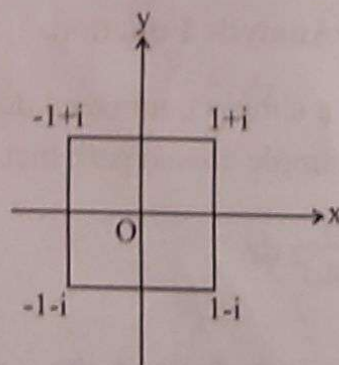
Thus, if $f(z)$ is analytic in D , then it has derivatives of all orders in D , and these derivatives are themselves analytic.

Examples

1. Evaluate $\int_C z^2 dz$, C : boundary of the square with vertices $1 + i$, $-1 + i$, $-1 - i$ and $1 - i$ (counter clockwise).

Here, z^2 is analytic everywhere, and therefore it is analytic on and inside C . By Cauchy's integral theorem,

$$\int_C z^2 dz = 0$$



2. Evaluate $\int_C e^{-z} dz$, C : unit circle $|z| = 1$, counter-clockwise.

Since e^{-z^2} is analytic everywhere, so by Cauchy's integral theorem $\int_c e^{-z^2} dz = 0$

3. Evaluate $\int_c \frac{1}{2z-1} dz$, $c : |z| = 1$

$$\text{Here, } \int_c \frac{1}{2z-1} dz = \frac{1}{2} \int_c \frac{1}{z-\frac{1}{2}} dz \quad (1)$$

The integrand is not analytic at $z = \frac{1}{2}$; and the point $z = \frac{1}{2}$ lies within the circle $|z| = 1$. So, we cannot use Cauchy integral theorem.

We know that

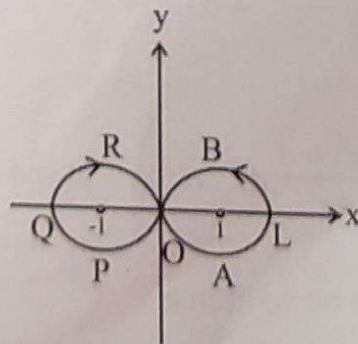
$$\int_c \frac{1}{z-z_0} dz = 2\pi i, \text{ where } c \text{ is any closed path containing}$$

z_0 in its interior.

$$\therefore \int_c \frac{1}{z-\frac{1}{2}} dz = 2\pi i$$

$$\text{From (1), } \int_c \frac{1}{2z-1} dz = \frac{1}{2} \cdot 2\pi i = \pi i$$

4. $\int_c \frac{dz}{z^2-1}$, where c is the contour given below



$$\text{Here, } \frac{1}{z^2 - 1} = \frac{1}{(z+1)(z-1)} = \frac{1}{2} \left(\frac{1}{z-1} - \frac{1}{z+1} \right)$$

$$\therefore \int_c \frac{dz}{z^2 - 1} = \frac{1}{2} \int_c \left(\frac{1}{z-1} - \frac{1}{z+1} \right) dz$$

$$= \frac{1}{2} \int_c \frac{1}{z-1} dz - \frac{1}{2} \int_c \frac{1}{z+1} dz \quad (1)$$

$$\text{Now, } \int_c \frac{1}{z-1} dz = \int_{\text{OALBOPQRO}} \frac{1}{z-1} dz$$

$$= \int_{\text{OALBO}} \frac{1}{z-1} dz + \int_{\text{OPQRO}} \frac{1}{z-1} dz$$

$$= 2\pi i + 0 = 2\pi i$$

$$\int_c \frac{1}{z+1} dz = \int_{\text{OALBO}} \frac{1}{z+1} dz + \int_{\text{OPQRO}} \frac{1}{z+1} dz$$

$$= 0 + (-2\pi i) = -2\pi i$$

Note: 0, because -1 lies outside OALBO
 $-2\pi i$, because integration is taken along OPQRO (clockwise)

$$\text{From (1), } \int_c \frac{dz}{z^2 - 1} = \frac{1}{2} \cdot 2\pi i - \frac{1}{2} (-2\pi i) = 2\pi i$$

5. Evaluate $\int_c \frac{2z^3 + z^2 + 4}{z^4 + 4z^2} dz$, $c : |z-2| = 4$ (clockwise)

The integrand can be written as

$$\begin{aligned}\frac{2z^3 + z^2 + 4}{z^4 + 4z^2} &= \frac{(z^2 + 4) + 2z^3}{z^2(z^2 + 4)} = \frac{1}{z^2} + \frac{2z}{z^2 + 4} \\ &= \frac{1}{z^2} + \frac{2z}{(z + 2i)(z - 2i)} \\ &= \frac{1}{z^2} + \frac{1}{z - 2i} + \frac{1}{z + 2i}\end{aligned}$$

The given integral reduces to

$$\int_c \frac{2z^3 + z^2 + 4}{z^4 + 4z^2} dz = \int_c \frac{1}{z^2} dz + \int_c \frac{1}{z - 2i} dz + \int_c \frac{1}{z + 2i} dz$$

We observe that points 0, 2i, -2i all lie within c.

$$\text{Here, } \int_c \frac{1}{z^2} dz = 0, \int_c \frac{1}{z - 2i} dz = -2\pi i \text{ (clockwise)}$$

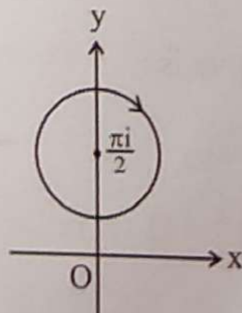
$$\int_c \frac{1}{z + 2i} dz = -2\pi i$$

$$\therefore \int_c \frac{2z^3 + z^2 + 4}{z^4 + 4z^2} dz = -4\pi i$$

$$6. \int_c \coth \frac{z}{2} dz \quad c : |z - \frac{\pi i}{2}| = 1, \text{ (clockwise)}$$

The integrand is not analytic when

$$\sinh \frac{z}{2} = 0 \quad \text{or} \quad \frac{e^{z/2} - e^{-z/2}}{2} = 0$$



Fundamentals of Engineering Mathematics-IV

$$\text{or, } e^z = 1 \quad \text{or } z = 2n\pi i \quad (n = 0, \pm 1, \pm 2, \dots)$$

$$\text{i.e. } z = 0, \pm 2\pi i, \pm 4\pi i, \dots$$

All these points lie outside the given circle and the integrand is analytic on and inside the circle. By Cauchy's integral theorem,

$$\int_C \coth \frac{z}{2} dz = 0$$

7. $\int_C \frac{z^3 \sin z}{3z-1} dz, \quad c : \text{unit circle, (counter clockwise)}$

The integrand is not analytic at the point where $3z-1=0$

It gives $z = \frac{1}{3}$, and this point lies within the unit circle.

$$\text{Now, } \int_C \frac{z^3 \sin z}{3z-1} dz = \int_C \frac{z^3 \sin z}{3\left(z - \frac{1}{3}\right)} dz = \int_C \frac{\frac{z^3 \sin z}{3}}{z - \frac{1}{3}} dz$$

$$\text{Let } f(z) = \frac{z^3 \sin z}{3} \quad \therefore f\left(\frac{1}{3}\right) = \frac{1}{3} \cdot \frac{1}{27} \sin \frac{1}{3} = \frac{1}{81} \sin \frac{1}{3}$$

$$\therefore \int_C \frac{z^3 \sin z}{3z-1} dz = 2\pi i \times f\left(\frac{1}{3}\right) = \frac{2}{81} \pi i \sin \frac{1}{3}$$

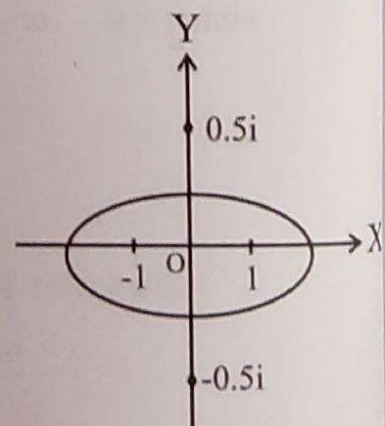
8. $\int_C \frac{z^2}{z^4-1} dz, \quad C : x^2 + 16y^4 = 4$

The equation of the curve is

$$x^2 + 16y^4 = 4$$

$$\text{or, } \frac{x^2}{4} + 4y^4 = 1$$

$$\text{or, } \frac{x^2}{4} + \frac{y^4}{1/4} = 1$$



The integrand is not analytic at the points where

$$z^4 - 1 = 0 \quad \text{or, } (z^2 + 1)(z^2 - 1) = 0$$

It gives $z = \pm i, z = \pm 1$. Of these singular points, only $z = 1$ and $z = -1$ lie within the curve (ellipse).

$$\begin{aligned} \text{Now, } \int_c \frac{z^2}{z^4 - 1} dz &= \int_c \frac{z^2}{(z^2 + 1)(z^2 - 1)} dz \\ &= \int_c \frac{z^2}{(z^2 + 1)(z - 1)(z + 1)} dz \\ &= \int_c \frac{z^2 / (z^2 + 1)}{(z - 1)(z + 1)} dz \quad (1) \end{aligned}$$

$$\text{Let } \frac{1}{(z - 1)(z + 1)} = \frac{A}{(z - 1)} + \frac{B}{(z + 1)}; \text{ then}$$

$$1 = A(z + 1) + B(z - 1)$$

$$\text{This gives } A = \frac{1}{2} \text{ and } B = -\frac{1}{2}$$

$$\therefore \frac{1}{(z - 1)(z + 1)} = \frac{1}{2} \left(\frac{1}{z - 1} - \frac{1}{z + 1} \right)$$

The integral on the right of (1), can be written as

$$\int_c \frac{z^2}{z^2 + 1} \frac{1}{2} \left(\frac{1}{z - 1} - \frac{1}{z + 1} \right) dz = \int_c \frac{z^2 / 2(z^2 + 1)}{z - 1} dz - \int_c \frac{z^2 / 2(z^2 + 1)}{z + 1} dz$$

$$\text{Let } f(z) = \frac{z^2}{2(z^2 + 1)} \quad \therefore f(1) = \frac{1}{4}, \quad f(-1) = \frac{1}{4}$$

$$\therefore \int_c \frac{z^2 / 2(z^2 + 1)}{z - 1} dz = 2\pi i \cdot f(1) = 2\pi i \cdot \frac{1}{4} = \frac{\pi i}{2}$$

$$\int_c \frac{z^2 / 2(z^2 + 1)}{z + 1} dz = 2\pi i \cdot f(-1) = 2\pi i \cdot \frac{1}{4} = \frac{\pi i}{2}$$

$$\therefore \int_c \frac{z^2}{z^4 - 1} dz = \frac{\pi i}{2} - \frac{\pi i}{2} = 0$$

9. $\int_c \frac{z^2 - z + 1}{z - 1} dz$, where c is the contour

(a) $|z| = 1$ (b) $|z| = \frac{1}{2}$

- a. The integrand is not analytic at $z = 1$; and this point lies on the circle $|z| = 1$.

Let $f(z) = z^2 - z + 1$; then $f(1) = 1 - 1 + 1 = 1$

$$\therefore \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times f(1) = 2\pi i \times 1 = 2\pi i$$

- b. Here, the point $z = 1$ lies outside the circle $|z| = \frac{1}{2}$; and so the integrand is analytic on and inside the circle $|z| = \frac{1}{2}$.

By Cauchy's integral theorem,

$$\int_c \frac{z^2 - z + 1}{z - 1} dz = 0$$

10. $\int_c \frac{\sin z}{z^4} dz$, $c: |z| = 1$

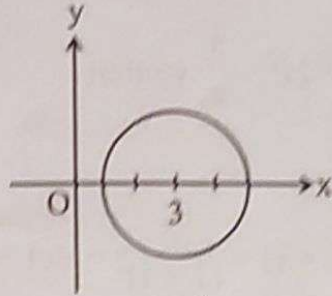
The integrand is not analytic at $z = 0$; and the point $z = 0$ lies within the circle $|z| = 1$.

Let $f(z) = \sin z$. Then $f^{(1)}(z) = \cos z$, $f^{(2)}(z) = -\sin z$, $f^{(3)}(z) = -\cos z$

$$\text{Now, } \int_c \frac{\sin z}{z^4} dz = \frac{2\pi i}{3!} f^{(3)}(0) = \frac{2\pi i}{3!} \times (-1) = -\frac{\pi i}{3}$$

11. $\int_c \frac{\ln z}{(z-2)^2} dz$, $c : |z-3|=2$

The integrand is not analytic at $z=0$ and $z=2$. The point $z=0$ lies outside the given circle; and the point $z=2$ lies within the circle.

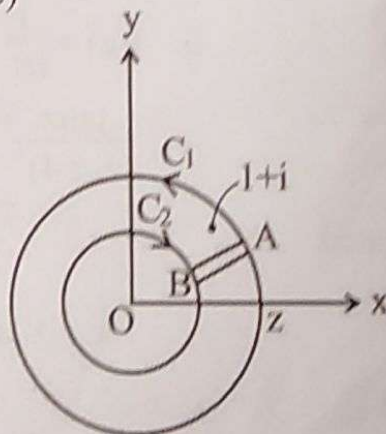


Let $f(z) = \ln z$ so that $f^{(1)}(z) = \frac{1}{z}$

$$\therefore f^{(1)}(2) = \frac{1}{2}$$

$$\text{Now, } \int_c \frac{\ln z}{(z-2)^2} dz = \frac{2\pi i}{1!} f^{(1)}(2) = \frac{2\pi i}{1} \times \frac{1}{2} = \pi i$$

12. $\int_c \frac{2z^3 - 3}{z(z-1-i)^2} dz$, c consists of $|z|=2$ (counter clockwise) and $|z|=1$ (clockwise)



The integrand is not analytic at the points $z=0$ and $z=1+i$. Make a cross-cut joining A and B. Then the region bounded by c_1 and c_2 is simply connected. Here, the point $z=1+i$ lies

within the region bounded by C_1 and C_2 but the point $z = 1+i$ does not lie. So, the integrand can be written as

$$\int_C \frac{2z^3 - 3}{(z - 1 - i)^2} dz$$

Let $f(z) = \frac{2z^3 - 3}{z} = 2z^2 - \frac{3}{z}$ so that

$$f^{(1)}(z) = 4z + \frac{3}{z^2}$$

$$\begin{aligned} \therefore f^{(1)}(1+i) &= 4(1+i) + \frac{3}{(1+i)^2} = 4(1+i) + \frac{3}{2i} \\ &= 4(1+i) - \frac{3}{2}i = \frac{8+5i}{2} \end{aligned}$$

$$\text{Now, } \int_C \frac{2z^3 - 3}{(z - 1 - i)^2} dz = 2\pi i \times f^{(1)}(1+i) = 2\pi i \left(\frac{8+5i}{2} \right) = \pi(8i - 5)$$

Exercise 4.2

1. Integrate the following functions counterclockwise around the unit circle.

i. $f(z) = \tan z/2$

ii. $f(z) = \frac{1}{|z|^2}$

iii. $f(z) = \frac{1}{z}$

iv. $\frac{\tan z}{(z - \pi/4)^3}$

2. Integrate as indicated

i. $\int_C \frac{dz}{z-a}$

$c : |z-a| = r$

ii. $\int_C \frac{z}{z^2 - 3z + 2} dz$

$c : |z+2| = 1/2$

iii. $\int_c \frac{2z^3 + z^2 + 4}{z^4 + 4z^2} dz$,

$c : |z-2| = 4$, (clockwise)

iv. $\int_c \frac{1}{z^2 + 1} dz$,

$c : (a) |z+i| = 1$ (b) $|z-i| = 1$

v. $\int_c \frac{z^2}{z^4 - 1} dz$,

$c : (a) |z+i| = 1$ (b) $|z| = 0.9$

vi. $\int_c \frac{2z + 1}{z^2 + z} dz$,

$c : |z| = \frac{1}{2}$

vii. $\int_c \tan z dz$,

$c : |z| = 1$

viii. $\int_c \frac{\cos \pi z}{z-1} dz$,

$c : |z| = 3$

ix. $\int_c \frac{4 - \sin z}{z^2 - 2z} dz$,

$c : \text{square with vertices } \pm 1 \text{ and } \pm i$

x. $\int_c \frac{e^{2z}}{(z+1)^4} dz$,

$c : |z| = 2$

xi. $\int_c \frac{z^6}{(2z-1)^6} dz$,

$c : |z| = 1$ (counter-clockwise)

xii. $\int_c \frac{\cot z}{(z-\pi/2)^2} dz$,

$c : 4x^2 + 9y^2 = 36$

xiii. $\int_c \frac{1}{(z^3 - 1)^2} dz$,

$c : |z - 1| = 1$

xiv. $\int_c \frac{z^4}{(z+1)(z-i)^2} dz$,

$c : 9x^2 + 4y^2 = 36$

xv. $\int_c \frac{(z+1)dz}{z^3 - 4z}$

c is given by

(a) $|z + 2| = \frac{3}{2}$ (b) $|z-2| = \frac{3}{2}$ (c) $|z| = \frac{3}{2}$

xvi. $\int_c \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz$ where c is any contour enclosing the point $z = -i$ (counter-clockwise)

- xvii. $\int_c \frac{1}{z^2 + 4} dz$, $c: 4x^2 + (y-2)^2 = 4$
 xviii. $\int_c \frac{\cosh 3z}{2z} dz$, $c: |z| = 1$
 xix. $\int_c \frac{z^3 - 6}{2z - i} dz$, $c: |z| = 2$
 xx. $\int_c \frac{z^3}{2z - i} dz$, $c: |z| = 1$
 xxi. $\int_c \ln(1-z) dz$, $c: \text{boundary of the parallelogram with vertices } \pm i, \pm(1+i)$
 xxii. $\int_c \frac{e^{5z}}{(z+i)^4} dz$, $c: |z| = 3$
 xxiii. $\int_c \frac{z^3 + \sin z}{(z-i)^3} dz$, $c: \text{the boundary of the square with vertices } \pm 2, \pm 2i$

Answers

1. (i) 0 (ii) 0 (iii) 0 (iv) $4\pi i$
 2. (i) $2\pi i$ (ii) 0 (iii) $-4\pi i$ (iv) $-\pi, \pi$
 (v) $-\pi/2, 0$ (vi) $2\pi i$ (vii) 0 (viii) $-2\pi i$
 (ix) $-4\pi i$ (x) $\frac{8\pi i e^{-2}}{3}$ (xi) $\frac{3\pi i}{32}$ (xii) $-2\pi i$
 (xiii) $\frac{-4\pi i}{9}$ (xiv) $4\pi(1-i)$
 (xv) (a) $-\frac{\pi i}{4}$ (b) $\frac{3\pi i}{4}$ (c) $-\frac{\pi i}{2}$
 (xvi) $-18\pi i$ (xvii) $\frac{\pi}{2}$ (xviii) πi
 (xix) $\frac{\pi}{8} - 6\pi i$ (xx) $\frac{\pi}{8}$ (xxi) 0
 (xxii) $\frac{125}{3} \pi i e^{-5i}$ (xxiii) $\pi i(6i - \sin i)$

4.9 Expansion of Analytic Function as Power Series

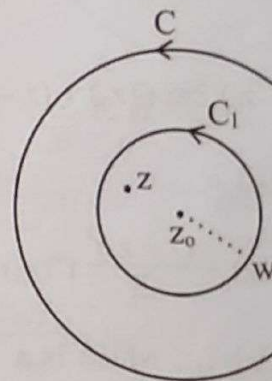
A Taylor series is a power series, and a power series represents an analytic function, and conversely every analytic function can be represented by a power series. Now we shall see how an analytic function can be represented by Taylor series.

Taylor Series

Let $f(z)$ be analytic inside a circle C with center z_0 . Then

$$f(z) = f(z_0) + (z - z_0)f'(z_0) + \frac{(z - z_0)^2}{2!}f''(z_0) + \dots$$

the series being convergent at every point inside C .



Let z be any point inside C , and let $|z - z_0| = r$. Let C_1 be a circle of radius ρ concentric with C such that $r < \rho$. By Cauchy's integral formula, we have

$$f(z) = \frac{1}{2\pi i} \int_{C_1} \frac{f(w)}{w - z} dw$$

$$\text{Now, } \frac{1}{w - z} = \frac{1}{w - z_0 - (z - z_0)} = \frac{1}{(w - z_0) \left(1 - \frac{z - z_0}{w - z_0}\right)}$$

$$= \frac{1}{w - z_0} \left(1 - \frac{z - z_0}{w - z_0}\right)^{-1}$$

4.9 Expansion of Analytic Function as Power Series

A Taylor series is a power series, and a power series represents an analytic function, and conversely every analytic function can be represented by a power series. Now we show that every analytic function can be represented by Taylor series.

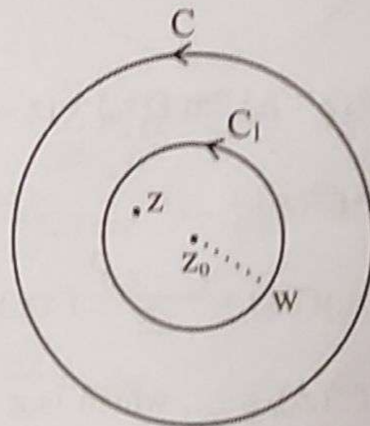
Taylor Series

Let $f(z)$ be analytic inside a circle C with centre z_0 and radius R . Then

$$f(z) = f(z_0) + (z - z_0)f'(z_0) + \frac{(z - z_0)^2}{2!}f''(z_0) + \dots + \frac{(z - z_0)^n}{n!}f^{(n)}(z_0) +$$

.....

the series being convergent at every point inside C .



Let z be any point inside C , and let $|z - z_0| = r$. Draw a circle C_1 of radius ρ concentric with C such that $r < \rho < R$. Let w be any point on C_1 . By Cauchy's integral formula, we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(w)}{w - z} dw$$

$$\begin{aligned} \text{Now, } \frac{1}{w - z} &= \frac{1}{w - z_0 - (z - z_0)} = \frac{1}{(w - z_0) \left(1 - \frac{z - z_0}{w - z_0}\right)} \\ &= \frac{1}{w - z_0} \left(1 - \frac{z - z_0}{w - z_0}\right)^{-1} \end{aligned}$$

$$= \frac{1}{w - z_0} \left[1 + \frac{z - z_0}{w - z_0} + \left(\frac{z - z_0}{w - z_0} \right)^2 + \dots + \left(\frac{z - z_0}{w - z_0} \right)^n + \dots \right]$$

Since $\left| \frac{z - z_0}{w - z_0} \right| < 1$, so the series is uniformly convergent. Hence, we can multiply by $f(w)$ and integrate term by term around C_1 . Multiplying by $f(w)$ and then integrating w.r.t. w over C_1 , we obtain

$$\int_{C_1} \frac{f(w)}{w - z} dw = \int_{C_1} \frac{f(w)}{w - z_0} dw + (z - z_0) \int_{C_1} \frac{f(w)}{(w - z_0)^2} dw + (z - z_0)^2 \int_{C_1} \frac{f(w)}{(w - z_0)^3} + \dots + (z - z_0)^n \int_{C_1} \frac{f(w)}{(w - z_0)^{n+1}} dw + \dots$$

Using Cauchy's integral formula and derivatives of analytic function, we have

$$2\pi i f(z) = 2\pi i f(z_0) + (z - z_0) 2\pi i f'(z_0) + (z - z_0)^2 \frac{2\pi i}{2!} f''(z_0) + \dots + (z - z_0)^n \frac{2\pi i}{n!} f^{(n)}(z_0) + \dots$$

$$\text{or, } f(z) = f(z_0) + (z - z_0) f'(z_0) + \frac{(z - z_0)^2}{2!} f''(z_0) + \dots + \frac{(z - z_0)^n}{n!} f^{(n)}(z_0) + \dots, \text{ which is a Taylor series and we}$$

say that $f(z)$ is expanded as a power series about $z = z_0$.

When $z_0 = 0$, then the above series reduces to

$$f(z) = f(0) + z f'(0) + \frac{z^2}{2!} f''(0) + \dots + \frac{z^n}{n!} f^{(n)}(0) + \dots,$$

which is known as **Maclaurin series**.

Laurent Series

Sometimes, we have to expand a function $f(z)$ around a point where $f(z)$ is not analytic. Taylor series fails in such a case. Here, we use Laurent series.

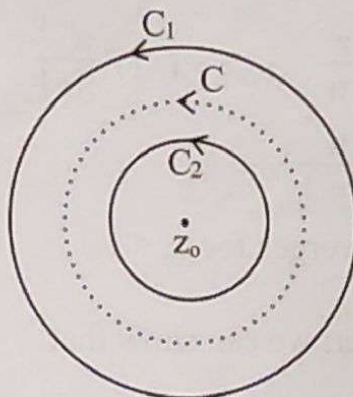
Laurent series is a series of non-negative integer powers of $z - z_0$ and negative integer powers of $z - z_0$.

The series of non-negative integer powers is called the **analytic part** and that of negative integer powers **principal part**.

Laurent series will be used for the classification of singularities.

Now we state **Laurent series** without proof.

If $f(z)$ is analytic in a ring-shaped region between two concentric circles C_1 and C_2 of radii r_1 and r_2 ($r_1 > r_2$) and on the circles themselves, then



$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n}, \quad \text{where}$$

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(w)}{(w - z_0)^{n+1}} dw \quad \text{and}$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(w)}{(w - z_0)^{-n+1}} dw$$

where C is any circle lying on the annulus.

Examples

1. Expand $\ln(1 + z)$ in a Taylor series about $z = 0$ and determine the region of convergence.

Let $f(z) = \ln(1 + z)$. Then

$$f'(z) = \frac{1}{1+z}, \quad f''(z) = -\frac{1}{(1+z)^2}, \quad f'''(z) = \frac{2!}{(1+z)^3}$$

$$\therefore f^{(n)}(z) = (-1)^{n-1} \frac{(n-1)!}{(1+z)^n}$$

Hence, $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2$,

$$f^{(n)}(0) = (-1)^{n-1} (n-1)!$$

The Taylor series about $z = 0$ is

$$f(z) = f(0) + zf'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots + \frac{z^n}{n!} f^{(n)}(0) + \dots$$

$$\ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots + (-1)^{n-1} \frac{z^n}{n} + \dots$$

$$\text{Here, } u_n = (-1)^{n-1} \frac{z^n}{n}, \quad u_{n+1} = (-1)^n \frac{z^{n+1}}{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} |z| = |z|$$

$\therefore$ The series converges for $|z| < 1$.

In a similar manner, we can show that

$$2. \quad e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots + \frac{z^{n-1}}{(n-1)!} + \dots$$

The series converges for all z , so the region of convergence is $|z| < \infty$.

$$3. \quad \frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots$$

The series converges for $|z| < 1$. The function $\frac{1}{1-z}$ is not analytic at $z = 1$; and the point $z = 1$ lies on the circle of convergence.

$$4. \quad \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

The series converges for $|z| < \infty$

$$5. \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots$$

The region of convergence is $|z| < \infty$

$$6. \cosh z = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \dots \quad (|z| < \infty)$$

$$7. \sinh z = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \dots \quad (|z| < \infty)$$

8. Find the Taylor series of the function $f(z) = \sin z$ with centre $\frac{\pi}{2}$. Also find the radius of convergence.

The function can be written as

$$f(z) = \sin\left(z - \frac{\pi}{2} + \frac{\pi}{2}\right) = \cos\left(z - \frac{\pi}{2}\right)$$

$$\text{Here, } f'(z) = -\sin\left(z - \frac{\pi}{2}\right), f''(z) = -\cos\left(z - \frac{\pi}{2}\right), f'''(z) = \sin\left(z - \frac{\pi}{2}\right)$$

$$f^{iv}(z) = \cos\left(z - \frac{\pi}{2}\right)$$

$$\text{Hence, } f\left(\frac{\pi}{2}\right) = 1, f'\left(\frac{\pi}{2}\right) = 0, f''\left(\frac{\pi}{2}\right) = -1, f'''\left(\frac{\pi}{2}\right) = 0,$$

$$f^{iv}\left(\frac{\pi}{2}\right) = 1$$

$$\begin{aligned} \text{Now, } f(z) &= f\left(\frac{\pi}{2}\right) + \left(z - \frac{\pi}{2}\right) f'\left(\frac{\pi}{2}\right) + \frac{\left(z - \frac{\pi}{2}\right)^2}{2!} f''\left(\frac{\pi}{2}\right) + \frac{\left(z - \frac{\pi}{2}\right)^3}{3!} \\ &\quad \times f'''\left(\frac{\pi}{2}\right) + \frac{\left(z - \frac{\pi}{2}\right)^4}{4!} f^{iv}\left(\frac{\pi}{2}\right) + \dots \\ f(z) &= 1 - \frac{\left(z - \frac{\pi}{2}\right)^2}{2!} + \frac{\left(z - \frac{\pi}{2}\right)^4}{4!} - \dots \end{aligned}$$

The series converges for all z ; so the radius of convergence is ∞ .

9. Find the Laurent series of $\frac{e^{z^2}}{z^3}$ with centre 0.

$$\text{Here, } \frac{e^{z^2}}{z^3} = \frac{1}{z^3} \left(1 + z^2 + \frac{z^4}{2!} + \frac{z^6}{3!} + \dots \right)$$

$$= \frac{1}{z^3} + \frac{1}{z} + \frac{1}{2}z + \frac{1}{6}z^3 + \dots, \text{ which is a Laurent series.}$$

The region of convergence is $|z| > 0$ or $0 < |z| < \infty$.

10. Obtain the Laurent series of $\frac{1}{z^2 + 1}$, $z_0 = i$ which converges for $0 < |z - z_0| < R$.

The given function can be written as

$$\begin{aligned} \frac{1}{z^2 + 1} &= \frac{1}{(z - i)(z + i)} = \frac{1}{(z - i)[(z - i) + 2i]} \\ &= \frac{1}{(z - i) 2i \left(1 + \frac{z - i}{2i}\right)} \\ &= \frac{1}{(z - i) 2i} \left(1 + \frac{z - i}{2i}\right)^{-1} \\ &= \frac{1}{(z - i) 2i} \left[1 - \frac{z - i}{2i} + \frac{(z - i)^2}{4i^2} - \frac{(z - i)^3}{8i^3} + \dots\right], \left(\left|\frac{z - i}{2i}\right| < 1\right) \\ &= -\frac{i}{2} (z - i)^{-1} + \frac{1}{4} + \frac{i}{8} (z - i) - \frac{1}{16} (z - i)^2 - \dots, \text{ which is a} \\ &\text{Laurent series.} \end{aligned}$$

$$\text{Now, } \left|\frac{z - i}{2i}\right| < 1 \Rightarrow |z - i| < 2$$

The series converges for $0 < |z - i| < 2$.

11. Obtain the Laurent series for $f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$ in the region $2 < |z| < 3$.

$$\text{Here, } f(z) = \frac{z^2 - 1}{z^2 + 5z + 6} = 1 - \frac{5z + 7}{(z + 2)(z + 3)} \quad (1)$$

$$\text{Let } \frac{5z + 7}{(z + 2)(z + 3)} = \frac{A}{z + 2} + \frac{B}{z + 3}, \text{ then}$$

$$5z + 7 = A(z + 3) + B(z + 2)$$

It implies that $A = -3$ and $B = 8$

$$\therefore \frac{5z + 7}{(z + 2)(z + 3)} = -\frac{3}{z + 2} + \frac{8}{z + 3}$$

Now (1) reduces to

$$\begin{aligned} f(z) &= 1 + \frac{3}{z+2} - \frac{8}{z+3} = 1 + \frac{3}{z\left(1+\frac{2}{z}\right)} - \frac{8}{3\left(1+\frac{z}{3}\right)} \\ &= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1} \\ &= 1 + \frac{3}{z}\left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \dots\right) - \frac{8}{3}\left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots\right) \end{aligned}$$

which is a Laurent series.

12. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series for the region $0 < |z+1| < 2$.

$$\begin{aligned} f(z) &= \frac{1}{(z+1)(z+3)} = \frac{1}{(z+1)[(z+1)+2]} \\ &= \frac{1}{(z+1) \cdot 2\left(1+\frac{z+1}{2}\right)} = \frac{1}{2(z+1)} \cdot \left(1+\frac{z+1}{2}\right)^{-1} \\ &= \frac{1}{2(z+1)} \left[1 - \frac{z+1}{2} + \frac{(z+1)^2}{4} - \frac{(z+1)^3}{8} + \dots \right] \\ &= \frac{1}{2(z+1)} - \frac{1}{4} + \frac{1}{8}(z+1) - \frac{1}{16}(z+1)^2 + \dots \end{aligned}$$

13. Obtain the Taylor series and Laurent series of the function

$$f(z) = \frac{1}{(z+2)(z^2+1)} \text{ when}$$

- (i) $|z| < 1$ (ii) $1 < |z| < 2$ (iii) $|z| > 2$

$$\text{Let } \frac{1}{(z+2)(z^2+1)} = \frac{A}{z+2} + \frac{Bz+C}{z^2+1}$$

$$\text{or, } 1 = A(z^2+1) + (z+2)(Bz+C)$$

$$\text{Put } z = -2, \text{ then } A = \frac{1}{5}$$

Comparing coefficients of z^2 and z , we get $B = -\frac{1}{5}$ and $C = \frac{2}{5}$

$$\therefore f(z) = \frac{1}{5} \left[\frac{1}{z+2} - \frac{z-2}{z^2+1} \right]$$

- i. For $|z| < 1$, the function $f(z)$ is analytic and so we have

$$\begin{aligned} f(z) &= \frac{1}{5} \left[\frac{1}{2 \left(1 + \frac{z}{2} \right)} - (z-2) (1+z^2)^{-1} \right] \\ &= \frac{1}{10} \left(1 + \frac{z}{2} \right)^{-1} - \frac{z-2}{5} (1+z^2)^{-1} \\ &= \frac{1}{10} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots \right) - \frac{z-2}{5} (1 - z^2 + z^4 - z^6 + \dots) \end{aligned}$$

The series being positive powers of z represents the Taylor series.

- ii. For $1 < |z| < 2$

$$\begin{aligned} f(z) &= \frac{1}{5} \left[\frac{1}{2 \left(1 + \frac{z}{2} \right)} - \frac{z-2}{z^2 \left(1 + \frac{1}{z^2} \right)} \right] \\ &= \frac{1}{10} \left(1 + \frac{z}{2} \right)^{-1} - \frac{z-2}{5z^2} \left(1 + \frac{1}{z^2} \right)^{-1} \\ &= \frac{1}{10} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots \right) - \frac{z-2}{5z^2} \left(1 - \frac{1}{z^2} + \frac{1}{z^4} - \frac{1}{z^6} + \dots \right), \end{aligned}$$

which represents the Laurent series.

- iii. For $|z| > 2$, we have

$$\begin{aligned} f(z) &= \frac{1}{5} \left[\frac{1}{z \left(1 + \frac{2}{z} \right)} - \frac{z-2}{z^2 \left(1 + \frac{1}{z^2} \right)} \right] \\ &= \frac{1}{5z} \left(1 + \frac{2}{z} \right)^{-1} - \frac{z-2}{5z^2} \left(1 + \frac{1}{z^2} \right)^{-1} \\ &= \frac{1}{5z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \dots \right) - \frac{z-2}{5z^2} \left(1 - \frac{1}{z^2} + \frac{1}{z^4} - \frac{1}{z^6} + \dots \right) \end{aligned}$$

which represents the Laurent series.

Exercise 4.3

- Expand the function $f(z) = \frac{1}{z}$ about $z=1$. Obtain the radius of convergence.
- Expand $f(z) = e^z$ about $z=a$.
- Expand $f(z) = \frac{1}{z^2 - 3z + 2}$ in the region
 - $|z| < 1$
 - $|z| > 2$
- Obtain the expansion of $f(z) = \frac{1}{z(1-z)}$.
- Expand $f(z) = \frac{1}{(z+i)^2}$ at $z=i$ by using the binomial expansion.
- Find the Laurent's series of $f(z) = \frac{z^2-4}{(z-1)^2}$ that converges for $0 < |z-1| < R$.
- Find the Taylor's and Laurent's series of the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$
 - for $|z| < 2$
 - for $|z| > 3$
 - for $2 < |z| < 3$
 - for $0 < |z+1| < 2$
 - for $|z| < 1$
- Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent series
 - $1 < |z| < 3$
 - $|z| > 3$
 - $0 < |z+1| < 2$
 - $|z| < 1$
- Find the Laurent's expansion for $f(z) = \frac{1}{z^2+1}$ given by
 - $0 < |z+1| < 1$
 - $1 < |z+1| < 2$
 - $|z+1| > 2$

Answers

- $f(z) = \frac{1}{2} - \frac{z-2}{4} + \frac{1}{8} (z-2)^2 - \dots$
- $e^z = e^a + \frac{(z-a)}{1!} e^a + \frac{(z-a)^2}{2!} e^a + \dots$
- a. $f(z) = \frac{1}{2} + \frac{3z}{4} + \frac{7z^2}{8} + \frac{15z^3}{16} + \dots$

Exercise 4.3

1. Expand the function $f(z) = \frac{1}{z}$ about $z = 2$ in Taylor's series.
Obtain the radius of convergence also.
2. Expand $f(z) = e^z$ about $z = a$
3. Expand $f(z) = \frac{1}{z^2 - 3z + 2}$ in the region
a. $|z| < 1$ b. $|z| > 2$
4. Obtain the expansion of $f(z) = \frac{1}{z(1-z)}$ in the region $|z+1| < \frac{3}{4}$
5. Expand $f(z) = \frac{1}{(z+i)^2}$ at $z = i$ by using Taylor's series.
6. Find the Laurent's series of $f(z) = \frac{e^z}{z - z^2}$
7. Find the expansion of $f(z) = \frac{z^2 - 4}{(z-1)^2}$ by using Laurent's series
that converges for $0 < |z-1| < R$
8. Find the Taylor's and Laurent's series which represents the
function $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$
(i) for $|z| < 2$ (ii) for $2 < |z| < 3$
(iii) for $|z| > 3$
9. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for
(i) $1 < |z| < 3$ (ii) $|z| > 3$
(iii) $0 < |z+1| < 2$ (iv) $|z| < 1$
10. Find the Laurent's expansion for $f(z) = \frac{7z - 2}{z^3 - z^2 - 2z}$ in the region
given by
(i) $0 < |z + 1| < 1$ (ii) $1 < |z + 1| < 3$
(iii) $|z + 1| > 3$

Answers

1. $f(z) = \frac{1}{2} - \frac{z-2}{4} + \frac{1}{8} (z-2)^2 - \dots + \frac{(-1)^n}{2^{n+1}} (z-2)^n + \dots$
2. $e^z = e^a + \frac{(z-a)}{1!} e^a + \frac{(z-a)^2}{2!} e^a + \dots$
3. a. $f(z) = \frac{1}{2} + \frac{3z}{4} + \frac{7z^2}{8} + \frac{15z^3}{16} + \dots$

$$b. f(z) = \frac{1}{z^2} + \frac{3}{z^3} + \frac{7}{z^4} + \dots$$

$$4. f(z) = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) (z+1)^n$$

$$5. f(z) = \sum_{n=0}^{\infty} \frac{(n+1)(-1)^n}{(2!)^{n+2}} (z-i)^n$$

$$6. f(z) = \frac{1}{z} \left[1 + 2z + \frac{5}{2}z^2 + \frac{17}{6}z^3 + \dots\right]$$

$$7. f(z) = 1 + \frac{2}{z-1} - \frac{3}{(z-1)^3}$$

$$8. (i) f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$$

$$(ii) f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$$

$$(iii) f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{z}\right)^n$$

$$9. (i) f(z) = -\frac{1}{2z^4} + \frac{1}{2z^3} - \frac{1}{2z^2} + \frac{1}{2z} - \frac{1}{6} + \frac{z}{18} - \dots$$

$$(ii) f(z) = \frac{1}{z^2} - \frac{4}{z^3} + \frac{13}{z^4} - \frac{50}{z^5} + \dots$$

$$(iii) f(z) = \frac{1}{z(1+z)} - \frac{1}{4} + \frac{(z+1)}{8} - \frac{(z+1)^2}{16} + \dots$$

$$(iv) f(z) = \frac{1}{3} - \frac{4}{9}z + \frac{13}{27}z^2 - \frac{80}{81}z^3 + \dots$$

$$10. (i) f(z) = -[1 + (z-1) + (z-1)^2 + (z-1)^3 + \dots] - \frac{3}{z+1} - \frac{2}{3} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{9} + \dots\right]$$

$$(ii) f(z) = \frac{1}{z+1} \left[1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \dots\right] - \frac{3}{z+1} - \frac{2}{3} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{9} + \dots\right]$$

$$(iii) f(z) = \frac{1}{z+1} \left[1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \dots\right] - \frac{3}{z+1} + \frac{2}{z+1} \left[1 + \frac{3}{z+1} + \frac{9}{(z+1)^2} + \dots\right]$$